1. Clinical Question & Core Hypothesis

Clinical Question

Why do some TNBC tumors that contain organized immune structures (TLS/TLS-like aggregates) still fail to respond to immune-based therapies — what tumor-intrinsic or microenvironmental factors modulate whether TLS confer functional antitumor immunity? 2) Tier 1 paths used: I built on the PD-L1 → immune-signaling path (connects checkpoint engagement and a pro/anti-tumor immune program) and the VEGF/Ktrans → serum VEGF path (connects angiogenic/vascular state to systemic VEGF as a modifier of immune access). These paths together explain how immune activation and vascular state could interact to determine immune effectiveness. 3) Tier 2 evidence supporting the claim: systemic inflammation (SII) and dynamic inflammatory biomarkers like CHI3L3/YKL-40 indicate host-tumor inflammatory tone that can further modulate TLS function; Ki-67 and CONUT provide proliferation and host-nutritional context that may alter immune competence; CCR5/NFKBII2 reflect immune signaling programs that shape TLS maturation. 4) Endpoint choice rationale: I selected therapy_response (immune-checkpoint therapy response as a therapy_response proxy) because the mechanistic question is explicitly about whether TLS translate into clinical benefit from immune therapies and how microenvironmental modifiers alter that effect — therapy response is the most direct, clinically actionable endpoint for this axis. 5) Deliberate exclusions: I excluded exploratory/underspecified signals (riboflavin, C-markers, generic lab values s0, ABSI) from the core hypothesis because they lack direct mechanistic links to TLS–immune–vascular interactions; they remain for exploratory analyses only?

Working Hypothesis

In TNBC, TLS-like lymphoid aggregates contribute to anti-tumor immunity only when two conditions co-occur: (1) high PD-L1 expression indicating antigen-engaged immune signaling, and (2) low VEGF-driven vascular dysfunction enabling vascular normalization and efficient T cell trafficking. Therefore TLS+PD-L1high/VEGFlow tumors will exhibit enhanced TLS maturation, greater intratumoral T cell activation, and higher response rates to PD-1/PD-L1 blockade compared with TLS+PD-L1low or VEGFlhigh contexts where VEGF-driven vascular abnormalities impede TLS function. This yields a testable three-way interaction: TLS presence × PD-L1 expression × VEGF level predicts therapy_response to ICIs in TNBC more accurately than any single marker alone. The hypothesis allows translation to a biomarker strategy: selecting patients with TLS-like aggregates who have PD-L1-high and VEGF-low tumors will identify those most likely to benefit from ICIs.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PD-L1, VEGF
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PD-L1: PD-L1 → immune-signaling path (connects checkpoint engagement and a pro/anti-tumor immune program) and the VEGF/Ktrans → serum VEGF path (connects angiogenic/vascular state to systemic VEGF as a modifier of immune access).

VEGF: VEGF/Ktrans → serum VEGF path (connects angiogenic/vascular state to systemic VEGF as a modifier of immune access).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	SII (systemic immune-inflammation index)	Supporting modifier from knowledge graph
Secondary	CHI3L3 (YKL-40)	Supporting modifier from knowledge graph
Secondary	CONUT score	Supporting modifier from knowledge graph
Secondary	Ki-67	Supporting modifier from knowledge graph
Secondary	CCR5	Supporting modifier from knowledge graph
Secondary	NFKBII2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Observational study in TNBC patients treated with ICIs to test main effects and interactions of TLS presence (imaging), PD-L1 expression (IHC), and VEGF (IHC or RNA), with endpoints including objective response rate (ORR), progression-free survival (PFS).

Prediction 2: Prospective biomarker cohort collecting pretreatment tumor biopsies and DCE-MRI vascular metrics (e.g., Ktrans) to quantify vascular normalization as a mediator between VEGF status and TLS function.

Prediction 3: Mechanistic preclinical work using murine TNBC models and humanized systems to manipulate VEGF signaling (e.g., anti-VEGF or VEGF overexpression) and PD-L1 pathways, assessing TLS maturation, T cell infiltration/activation, and response to ICIs.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SII (systemic immune-inflammation index): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CHI3L3 (YKL-40): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.